SFA1500A

Laser Rangefinder Module

Product Manual

Version 202606V1

Disclaimer

This product manual is intended to provide reference specifications and performance parameters of the product. If certain specifications are not included in this document, please contact SunFlaser Technology (Chengdu) Co., Ltd. directly for further confirmation. All contents of this manual and the specifications of the ranging module are subject to change without prior notice.

Without the prior written consent of SunFlaser Technology (Chengdu) Co., Ltd., this document may not be reproduced, stored in any retrieval system, transmitted, or used for any unauthorized purpose in any form or by any means. This document may not be disclosed or distributed to any third party without the explicit authorization of SunFlaser Technology (Chengdu) Co., Ltd.

Due to variations in actual product performance under different operating conditions, the specifications and parameters provided in this document are for reference purposes only and do not constitute any form of warranty. Any express or implied warranties, including but not limited to warranties of merchantability or fitness for a particular purpose, shall be subject to the actual performance of the specific product.

SunFlaser Technology (Chengdu) Co., Ltd. reserves the right to revise or update this document at any time without prior notice. This document may contain technical inaccuracies or typographical errors. Feedback and suggestions are welcome, and verified issues will be corrected in subsequent revisions of the manual.

At the time of publication, all descriptions, specifications, designs, and related procedures contained herein are considered valid. SunFlaser Technology (Chengdu) Co., Ltd. reserves the right to modify the above information at any time without prior notice. If you have any questions regarding the use of this manual, please contact us promptly.

Safety Instructions

For the safe use of the ranging module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not tamper with or attempt to alter the performance of the ranging module.

Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the ranging module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the ranging module.
- Operate the ranging module only within the specified voltage and power range.
- Operate the ranging module in a clean, dry, and ESD-protected environment.
- Do not touch the optical glass lens with your fingers.
- Avoid exposing the ranging module to environments with drastic temperature fluctuations.

Improper Use

- Using the ranging module with the host system before verifying that all functions are operating properly;
- Operating the ranging module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the ranging module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the ranging module in non-standard, abnormal, or unsuitable environments;

- Operating the ranging module in a manner inconsistent with the instructions provided in this manual;
- Continuing to operate the ranging module after improper handling or misuse;
- Using a ranging module with obvious faults, defects, or damage;
- Performing test operations within the blind zone of the ranging module.

Improper use may result in the following risks:

- Eye injury
- Incorrect results
- Measurement errors
- Property damage
- Equipment malfunction

Operator Responsibilities

- Be familiar with the basic knowledge required for the correct operation of the ranging module;
- Understand and comply with the operation and maintenance requirements of the ranging module;
- Prevent damage to the ranging module during storage and transportation;
- Keep the ranging module clean and well maintained;
- Ensure that the ranging module is stored and operated in a safe environment;
- Do not expose the ranging module to strong mechanical shock or impact;
- Do not expose the ranging module to environments with drastic temperature fluctuations.

Avoid Measurement Errors

- Pay attention to all factors that may affect the performance of the ranging module;

- After the laser rangefinder module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the ranging module.

Table of Contents

01 Product Overview

The SFA1500A is a highly integrated miniature laser rangefinder module based on a 905 nm laser ranging solution. Featuring a compact, lightweight, and low-power design, it is ideal for platforms with strict space and payload constraints.

The module supports multiple ranging modes and features a simple, universal communication interface, enabling rapid integration with UAVs, electro-optical gimbals, intelligent robots, handheld devices, and various embedded electro-optical platforms.

With excellent environmental durability, stable and reliable ranging performance, and excellent anti-interference capability, the SFA1500A is well suited for target ranging, environmental sensing, and system integration applications in complex outdoor operating conditions.

02 Key Features

Precision Laser Ranging

Equipped with a 905 nm laser ranging solution, the module provides stable long-range distance measurement with high accuracy and strong resistance to environmental interference, making it suitable for target detection in various complex outdoor scenarios.

Multiple Ranging Modes

Supports single ranging and multi-frequency continuous ranging modes, with the ability to stop ranging operations at any time, enabling flexible adaptation to different device operating cycles and application requirements.

Communication and System Integration

Features a standard TTL serial communication interface with a simple and universal protocol, allowing rapid integration with UAV flight controllers, electro-optical gimbals, embedded controllers, and various intelligent systems for custom system integration.

High Reliability and Anti-Interference Performance

The module features low false alarm rates, high measurement reliability, a wide operating temperature range, and a durable vibration-resistant structure, ensuring long-term stable operation in airborne, portable, and harsh industrial outdoor environments.

Optimized for Lightweight Integration

Adopting a compact modular design with small size, light weight, and low power consumption, the module is ideal for embedded installation in compact intelligent devices with strict space and payload limitations.

03

Key Specifications

No.	Item	Specification
1	Operating Wavelength	905 nm ± 10 nm
2	Transmitting Aperture	Ø 7.2 mm
3	Receiving Aperture	Ø 19 mm
4	Baud Rate	115200 bps
5	Ranging Capability	≥1500 m (Large Target)
6	Minimum Measuring Distance	≤10 m
7	Ranging Accuracy	±1 m (Distance < 300 m) ±2 m (Distance ≥ 300 m)
8	Measurement Success Rate	≥98%
9	False Alarm Rate	≤2%
10	Ranging Frequency	Single-shot, 1 Hz, 2 Hz
11	Weight	≤20 g
12	Dimensions	≤ Ø23 mm × 46 mm
13	Electrical Interface	Receptacle: A1002WR-S-4P LNN100-157128-200-4P — Plug: LNN100-157128-200-4P
14	Supply Voltage	3 V–5 V (Typical: DC 3.3 V)
15	Standby Power Consumption	≤0.4 W @ 3.3 V
16	Average Power Consumption	≤0.5 W @ 3.3 V
17	Peak Power Consumption	≤1.2 W @ 3.3 V
18	Operating Temperature	-20 °C to +60 °C
19	Storage Temperature	-30 °C to +60 °C

- Note 1: Parameters marked with an asterisk (*) in the table are design values or theoretical calculation values and are provided for reference only.
-

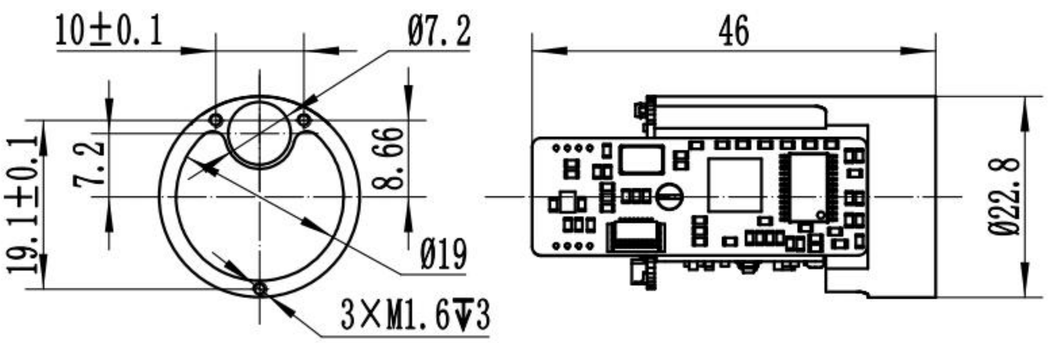
- Note 2: The nominal ranging distance and accuracy are achieved under environmental conditions with visibility not less than 10 km and relative humidity not greater than 60%. The ranging capability and accuracy may vary depending on environmental conditions and target characteristics; actual performance shall be subject to field measurements. When the rangefinder is mounted on a stabilized platform, ensure that the platform can maintain stable target tracking; otherwise, the ranging distance and accuracy may be affected.

04

Interface and Connection

Mechanical and Optical Interface

- The mechanical and optical interface dimensions of the laser rangefinder module are shown in the figure below. Unit: mm.



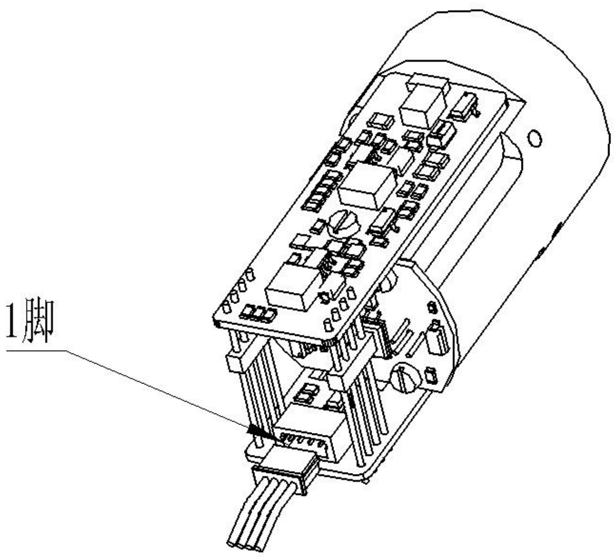
1、单位：mm；

2、未注公差按GB/T1804-2000m执行。

Electrical Interface:

- The electrical interface requirements are as follows:
  - Supply Voltage: 3 V–5.5 V (Typical: 3.3 V)
  - Standby Power Consumption: ≤0.4 W
  - Average Power Consumption: ≤ 0.5 W
- The host-side interface is connected to the rangefinder-side connector (A1002WR-S-4P) through an LNN100-157128-200-4P connector for integration testing. The pin definitions are shown in the table below, and the position of Pin 1 on the connector is illustrated in the figure below.

Pin Number	Label	Electrical Characteristics Definition	Signal Direction
P-1	VIN+	Positive Power Input	Power Supply
P-2	VIN-	Negative Power Input	
P-3	TTL_RXD	Signal Input Port	Host to Rangefinder
P-4	TTL_TXD	Signal Output Port	Rangefinder to Host



05

Communication Protocol and Commands

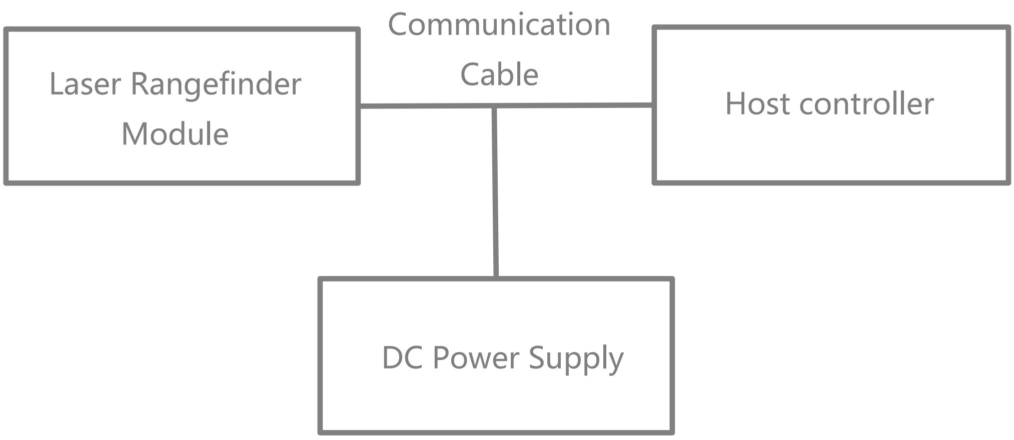
For details of the following contents, please refer to Appendix 1:

- Communication Interface
- Single Measurement Command
- Continuous Measurement Command
- Stop Measurement Command

06 Operating Instructions

Power-On Procedure

- Before powering on, connect the laser rangefinder module, communication cable, DC power supply, and host computer according to the electrical interface requirements and as shown in the figure below.



- The module powers up automatically after the power supply is connected.

Power-Off Procedure

- Before powering off, ensure that all operating processes and tasks have been completed and that the program has been exited properly.
- After confirmation, disconnect the power supply to power off the module normally.

Ranging Mode Operation

- Send the "Single Ranging" command to the laser rangefinder module. The module performs a single ranging operation and returns the ranging status and distance value.
- Send the "Continuous Ranging" command to the laser rangefinder module. The module performs continuous ranging and returns the ranging status and distance value continuously.
- Send the "Stop Ranging" command to terminate the ranging operation.

07 Precautions for Use

- Direct eye exposure to the rangefinder laser should be avoided. If observation is necessary, appropriate professional protective equipment must be used. Never stare directly into the laser beam, as it may cause eye injury.
- Avoid ranging targets within the minimum measuring distance, especially highly reflective objects such as glass or polished metal surfaces. Avoid operating multiple rangefinder modules face-to-face at close range. Prevent high-energy laser sources from directly illuminating the receiving aperture of the module. During assembly and debugging, the receiving lens should remain covered at all times; otherwise, excessively strong return signals may permanently damage the detector.
- Observe electrostatic discharge (ESD) protection precautions. The electronic components inside the laser rangefinder are sensitive to static electricity. Do not touch any electronic components without appropriate protective measures.
- Do not touch the optical lenses directly with fingers or hard objects. Avoid using cleaning agents on the lenses unless absolutely necessary, as improper cleaning may cause irreversible damage to the optical coating and degrade product performance.
- Do not operate the product under direct sunlight. Do not store the product in highly polluted environments or outside the specified storage temperature range. Avoid exposing the laser rangefinder to severe mechanical stress such as vibration, shock, or dropping. Rapid temperature changes should also be avoided, as they may adversely affect product performance.
- Although the rangefinder incorporates a certain level of sealing protection, it is recommended to operate the product in environments with humidity below 80% and to maintain a clean operating environment in order to extend service life.

- Do not attempt to disassemble, modify, or repair any part of the rangefinder. Unauthorized adjustment or alteration of product performance is strictly prohibited.
- Operate the rangefinder strictly in accordance with the instructions provided in this manual. Do not send commands other than those specified in the communication protocol.

08 Optical Window Requirements

- Material Recommendations

The optical window material is recommended to be Chengdu Guangming Optical Glass H-K9L or fused silica. H-K9L and fused silica are commonly used colorless optical glasses suitable for laser wavelengths from 300 nm to 2100 nm. They offer high cost-effectiveness and excellent physical properties.

- Manufacturing Recommendations

The wedge angle tolerance of the optical window should be as small as possible. It is recommended that the wedge angle tolerance be ≤ 3′ (tolerance grade ≤ 7). The optical surface of the window should be as smooth as possible, with a recommended arithmetic average roughness (Ra) of 0.012 μm.

> Coating Recommendations:

For 905 nm laser rangefinder modules, it is recommended to apply an anti-reflection (AR) coating optimized for the 855 nm–955 nm wavelength range, with a transmission rate of ≥99.5%.

Depending on the specific application environment, additional protective coatings such as hydrophobic coatings or hard coatings may be applied to the outer surface of the optical window. Other performance requirements should comply with GJB 2485-95, with a transmission rate of ≥98%.

> Optical Window Shape and Usage Recommendations:

The effective aperture of the optical window varies depending on different products. Its overall dimensions should satisfy the following requirements: the difference between the optical window effective aperture and the outer diameter should be ≥ 2 mm; and the difference between the rangefinder antenna outer diameter and the projected size of the optical window effective aperture should be ≥ 1.5 mm, as shown in the schematic diagram below.

Since the optical window has a certain level of laser absorption, it is recommended that the thickness of the optical window be controlled within 2–4 mm according to the overall size.

Due to the high transmittance of the optical window, it is recommended that the emission optical axis be aligned parallel to the normal of the optical window surface. The positioning relationship between the optical window and the dual optical barrels is shown in the schematic diagram below.

Meanwhile, the air gap between the optical window and the rangefinder should be less than 0.5 mm. The figure below illustrates two installation configurations of the optical window.

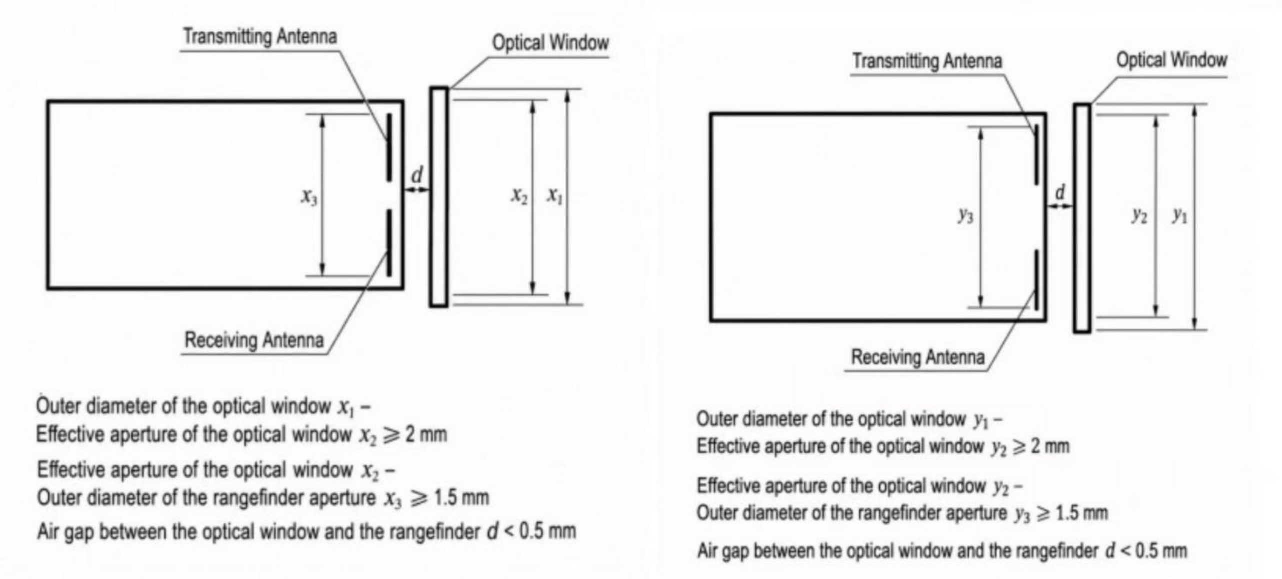


Figure 1: Optical Window Dimensions and Placement Diagram

09 Safety Precautions

- To ensure the safety of both operators and test subjects, the following safety measures have been incorporated into the design of the SFA1500A laser rangefinder module;
- Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;
- Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- The module operates below the safety voltage limit for the human body.

10 Maintenance Instructions

Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

No.	Item	Quantity	Remarks
1	Laser Rangefinder Module	1 unit	-
2	Certificate of Conformity	1 copy	-
3	Packaging Box	1 pc	-
4	Connector Plug	1 pc	-

Module Cleaning

Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the ranging module, connectors, and cables away from moisture and contaminants whenever possible;
- Ensure that the equipment is completely dry before packaging.

Inspection and Maintenance

Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure

- Check whether the product appearance is normal;

- Check whether the cable connections are correct and securely connected.

- Power-On Inspection Procedure
  - Perform the power-on procedure and verify that the standby power is correct;
  - Start the self-test function of the product;
  - After the inspection is completed, perform the power-off procedure.

Periodic Maintenance

Under normal operating conditions, the laser rangefinder module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)
  - All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
  - All screws on the panel shall be securely fastened;
  - The optical glass of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.
- Comprehensive Power-On Inspection and Maintenance
  - Connect the product power supply in sequence;
  - Perform the power-on procedure and verify that the standby power is correct;
  - Start the product self-test and complete the self-test procedure;
  - Perform the power-off procedure after completion of the inspection.

Troubleshooting

The following list only includes possible fault phenomena not caused by the product itself. If any other technical problems occur during operation, please contact our company for technical support.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
No communication or abnormal communication of the rangefinder module	Power supply issue	Check whether the power supply has hardware or software wake-up enabled. Verify that the voltage setting is correct and ensure the protection current is properly configured. If the voltage is correct, try replacing the power supply and test again.	If the module operates normally after changing the voltage or power supply, the issue can be determined to be caused by an incorrect power supply.	Set the correct voltage and current, or replace the power supply.
	Connection cable issue	Check whether the wiring is secure, or replace the communication cable for testing (TTL communication cables should not be excessively long).	If communication is restored after rewiring or replacing the communication cable, the issue can be determined to be caused by a connection cable problem.	Rewire the connections or replace the communication cable.
		Check whether the wiring matches the interface pin definition.	If the module operates normally after confirming that the wiring matches the pin definition, the issue can be determined to be caused by incorrect wiring.	Reconnect the wiring according to the correct pin definition.
	Incorrect baud rate setting	1. Check whether the rangefinder status and baud rate settings are consistent with the communication protocol. 2. If the rangefinder supports baud rate configuration, change to the required baud rate and try communication again.	If communication operates normally after aligning the baud rate with the communication protocol, the issue can be determined to be caused by an incorrect baud rate setting.	Set the host computer baud rate to match the communication protocol.
	Incorrect communication command	Check the rangefinder status and verify whether the transmitted command is consistent with the communication protocol, and confirm that the command is in hexadecimal format.	If communication operates normally after confirming that the transmitted command matches the communication protocol, the issue can be determined to be caused by an incorrect communication command.	Send the correct command to the rangefinder module.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	RS422 driver not installed on the host computer.	Install the RS422 driver on the host computer.	If communication with the rangefinder module works normally after installation, the issue can be determined to be caused by the missing driver installation.	Install the driver.
	Main control program issue	If communication fails during integration with the main controller, first check whether standalone communication is functioning normally.	If standalone communication works normally, the issue can be considered likely to be caused by a problem in the main control program.	It is recommended to check whether there are issues in the system-level program (e.g., whether the parity/check bits are correct, and whether the host computer data parsing and display are implemented correctly).
	Other causes (e.g., circuit failure, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
The rangefinder returns invalid values.	Operational error / improper operation.	Check whether the receiving or transmitting lens is blocked, and ensure that there are no obstructions in the measurement environment (e.g., window glass, wall corners, etc.).	If the rangefinder returns normal readings after removing the obstruction, the issue can be considered likely caused by optical blockage.	Remove the obstruction.
		Whether the ranging axis is aligned with the optical sighting axis.	If the rangefinder returns normal measurements after realignment, the issue can be determined to be caused by misalignment between the ranging axis and the optical sighting axis.	Realign the ranging and optical axes and perform measurement again.

<table>
<tr><th>Problem Description</th><th>Possible Causes of Failure</th><th>Troubleshooting Method</th><th>Evaluation Criteria</th><th>Solution / Corrective Action</th></tr>
<tr><td rowspan="2"></td><td>Laser transmitter failure.</td><td>Check whether the current reading is normal. Observe whether the current fluctuates during ranging. Send a self-test command to check the returned data. Alternatively, use an infrared test card or a CCD camera to verify whether a laser spot is present.</td><td>If the emission self-test is abnormal and no laser spot is observed, the issue can be preliminarily determined to be caused by a laser transmitter failure.</td><td>Return to factory for repair.</td></tr>
<tr><td>Other causes (e.g., circuit failure, detector, etc.)</td><td>Return to factory for inspection and troubleshooting.</td><td>Return to the factory for inspection and diagnosis.</td><td>Return to factory for repair.</td></tr>
<tr><td rowspan="3">False ranging results.</td><td>Electrical noise interference.</td><td rowspan="3">Block or point the rangefinder receiving lens toward open space (or away from the sun) and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by optical noise interference. Further determine whether it is due to external ambient light or the system optical window (e.g., mismatch of wavelength band with the rangefinder, excessive tilt angle of the optical window, small aperture causing light blockage, or a contaminated optical window).
Power the rangefinder module independently within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by electrical or communication noise interference.
Power the rangefinder module independently within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by power supply noise interference.
Operate the rangefinder module with communication only within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by communication noise interference.
If false alarms still occur after the above steps, test the rangefinder module as a standalone unit (removed from the system and powered/communicated independently). If false alarms are still present, the issue can be determined to be caused by the rangefinder module itself or the power adapter.</td><td rowspan="3"></td><td>It is recommended to address power supply noise issues by adding a larger common-mode filter in series, replacing the grounding capacitor with a 10 nF capacitor, and increasing the detection threshold (note that this may reduce system performance).</td></tr>
<tr><td>Communication noise interference.</td><td>It is recommended to address communication noise interference by adding a ferrite core (shielding magnetic ring) to the cable or increasing the threshold level (note that this may reduce system performance).</td></tr>
<tr><td>Optical noise interference.</td><td>It is recommended to address optical noise issues (optical window condition and test angle) and increase the detection threshold (note that this may reduce system performance).</td></tr>
</table>

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Other causes (e.g., circuit failure, etc.).	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
Insufficient performance capability of the rangefinder module.	Environmental conditions (weather-related factors).	Check whether the weather conditions include rain, snow, fog, or dust/sand, whether atmospheric visibility is too low, whether humidity is too high, or whether there are localized atmospheric turbulence effects, as well as any discrepancies between the environmental conditions and the test site.	If the ranging performance meets the technical specifications when the visibility is above the required threshold, the issue can be determined to be caused by environmental weather conditions.	Test the rangefinder performance under appropriate weather conditions.
	Optical window issue.	Determine, by checking the system configuration, whether the issue is caused by low optical window transmittance, incorrect wavelength band, contamination, small aperture causing light blocking, or excessive tilt angle. Remove the optical window and test the rangefinder module independently to observe whether its ranging performance is degraded. Also inspect whether the lens of the rangefinder is excessively contaminated.	If the ranging performance meets the specification requirements after removing the optical window, the issue can be determined to be caused by low transmittance of the host optical window. If the ranging performance meets the specification requirements after cleaning the rangefinder lens, the issue can be determined to be caused by low transmittance of the host optical window.	Replace the optical window according to the recommended selection and installation guidelines, or switch to a higher-performance rangefinder model. Clean the lens.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Target size is too small or target reflectivity is too low.	Replace with another target. It is recommended to use targets such as buildings that are perpendicular to the optical axis, and avoid hollow or perforated targets as much as possible.	If the ranging performance meets the specification requirements after replacing with a higher-reflectivity or larger target, the issue can be determined to be caused by inappropriate target selection.	Replace the target, or switch to a higher-performance rangefinder model.
	Operating temperature exceeds the specified range of the rangefinder module.	When ranging performance degrades, send a self-test command to check the current operating temperature of the rangefinder module. After the temperature returns to normal, retest and observe whether the performance issue persists.	If the operating temperature exceeds the specified range when performance degrades, and the performance issue disappears after the temperature returns to normal, the issue can be determined to be caused by the operating environment exceeding the allowable temperature range of the rangefinder module.	Operate the device within the specified operating temperature range.
	Optical axis misalignment.	Check whether high/low temperature tests, thermal shock tests, shock tests, or vibration tests were performed prior to the degradation of ranging performance, or whether any drop or impact event occurred.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
	Other causes (e.g., circuit failure, low receiver sensitivity, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
Accuracy out of tolerance.	Large variation in target reflectivity.	Confirm whether the tested target is a high-reflectivity target, and determine whether the issue is large data fluctuation or accuracy out of tolerance.	If the tested target is highly reflective and the ranging accuracy returns to normal after replacing it with a suitable target, the issue can be determined to be caused by excessive target reflectivity variation.	Replace the target. The use of highly reflective test targets is not recommended.
	Test operation issue.	Check whether the target has been properly aimed at.	If the ranging accuracy returns to normal after correctly aiming at the target, the issue can be determined to be caused by improper test operation.	Re-aim at the target.
	Instrument issue.	Verify the accuracy of the calibration instrument (possible calibration error).	If the ranging accuracy returns to normal after replacing with an accurately calibrated instrument, the issue can be determined to be caused by the test instrument.	Replace the calibration instrument.
	Low optical window transmittance.	Check whether the ranging value is abnormally high.	If the ranging accuracy returns to normal after removing the optical window, the issue can be determined to be caused by the optical window.	Replace the optical window according to the recommended optical window selection and installation guidelines.

Packaging and Transportation Requirements

Packaging

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

Transportation

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

11 Technical Support and Service

If a product malfunction occurs and factory return is required for fault analysis, troubleshooting, or repair, SunFlaser Technology (Chengdu) Co., Ltd. provides a one-year warranty and lifetime technical support from the date of product delivery.

During the warranty period, product failures caused by manufacturing or quality issues will be repaired or replaced free of charge by SunFlaser. For product damage caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and operate it carefully during use. If you have any additional questions regarding operation or maintenance, please contact our after-sales support personnel at any time.

12 Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1:

1. 1 Data

- Baud Rate: 115200 bps;
- Single-byte transmission format: 1 start bit, 8 data bits, no parity bit, and 1 stop bit. For 8-bit data transmission, the low byte is transmitted first, followed by the high byte.

1. 2 Communication Protocol I

1.2.1 Single Ranging Command

- Note: Transmit checksum = Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7;
- Receive checksum = Byte 1 + Byte 2 + Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7.

Transmit to the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x88	0xFF	0xFF	0xFF	0xFF	Checksum

Response from the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x88	status	0xFF	DATA_H	DATA_L	Checksum

- Status = 0: Single ranging failed; DATA_H = 0xFF, DATA_L = 0xFF.
- Status = 1: Single ranging successful; DATA_H = High byte of the measurement result; DATA_L = Low byte of the measurement result.

1.2.2 Continuous Ranging Command

- Note: Transmit checksum = Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7;
- Receive checksum = Byte 1 + Byte 2 + Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7.

Transmit to the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	Freq	0xFF	0xFF	0xFF	0xFF	Checksum

Response from the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	Freq	status	0xFF	DATA_H	DATA_L	Checksum

- Status = 0: Continuous ranging failed; DATA_H = 0xFF, DATA_L = 0xFF.
- Status = 1: Continuous ranging successful; DATA_H = High byte of the measurement result; DATA_L = Low byte of the measurement result.

- Freq = 0x89: 1 Hz ranging;
- Freq = 0xA9: 2 Hz ranging;
- Freq = 0xB9: 5 Hz ranging;
- Freq = 0xF9: Alignment mode (returns alignment status once after receiving the alignment command).

1.2.3 Stop Ranging

Transmit to the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x8E	0xFF	0xFF	0xFF	0xFF	Checksum

Response from the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x8E	status	0xFF	0xFF	0xFF	Checksum

- Status = 0: Failed to stop continuous ranging;
- Status = 1: Continuous ranging stopped successfully.
- Note: Returned data is transmitted in hexadecimal format. All measurement results are output as the actual value multiplied by 10.

- Example: If dist = 2000.3 m, the output data is 20003. Converted to hexadecimal, the value is 0x4E23, where Data1 = 0x4E and Data2 = 0x23.

1.3 Communication Protocol II

1.3.1 Information

- The control command message format is shown in Table 1.

Table 1 Control Command Messages Received by the Rangefinder

Byte	Description	Byte Data (Header, Command, Tail)	Remarks
1	Header	0x55	
2	Command Code	0x02	Single Ranging
		0x03	1 Hz Ranging
		0x06	2 Hz Ranging
		0x04	5 Hz Ranging
		0x00	Stop Ranging
		0x0A	Alignment Command
3	Tail	0xAA	

1.3.2 Rangefinder Return Data

- The rangefinder return data is shown in Table 2.

Table 2 Rangefinder Return Data

Byte	Description	Byte Value (Hexadecimal)
1	Header	0xAA
2	High Byte of Integer Part of Target Distance	
3	Low Byte of Integer Part of Target Distance	
4	Decimal Part of Target Distance	0x00 to 0x63
5	Tail	0x55

Email:sales@sunflaser.com www.sunflaser.com